DETAILED MODELLING OF THE EXPANSION REACTIVITY FEEDBACK IN FAST REACTORS USING OpenFOAM

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ABSTRACT

The correct prediction of expansion-related reactivity feedbacks is paramount to the safety analysis of fast reactors. Traditionally, these feedbacks were evaluated via uniform radial and axial expansions based on average diagrid and fuel temperatures, respectively. The resulting smearing of the neutronic effects entails some limitations in terms of representation of local effects and overall accuracy. In addition, the thermo-mechanical behavior of a core is a complex phenomenon that cannot be reduced to simple uniform axial and radial expansions. Common examples of complex deformation patterns are the core compaction and the core flowering, which are known to have potentially dramatic impacts on core safety. In this paper, a model is developed that allows for a detailed thermo-mechanical analysis of the core, and that is capable of using this information for a locally detailed deformation of the geometry employed in the neutronic model. The capabilities of the model are discussed, together with its present limitations and some needs for improvements.

KEYWORDS: Core deformation, fast reactors, reactivity feedbacks, OpenFOAM, GeN-Foam

1. INTRODUCTION

Thermal expansions represent one of the main mechanisms for negative reactivity feedbacks in fast reactors and are essential for compensating the generally positive (at least locally) coolant temperature feedback. The effect of thermal expansions can broadly be split into three main effects:

- A local change of fuel-to-coolant ratio. This provides a negative contribution to reactivity and often represents the dominant effect.
- A variation of core boundaries. This is often a second-order effect and it provides a positive reactivity contribution.
- Connected to the second effect, a change in the relative position of core and control rods. This effect strongly depends on the core design and it can be of primary importance, notably in slow transients. It is typically a strongly negative effect.

Prediction of these effects is traditionally done via a uniform radial and axial expansion of the core based on average diagrid and fuel temperature, respectively. In deterministic codes, the multi-group cross-sections are then parametrized based on the average expansion of fuel and diagrid, which allows to reproduce the spectral effect of a changing fuel-to-coolant ratio. The change in the relative position of core and control rods is typically achieved by modifying the control rod position based on the temperature of the control-rod driveline. An example of this state-of-the-art treatment of reactivity effects is the FAST code system employed at the PSI [1].

Since a few years some research activities have started exploring the possibility of creating new solvers for reactor transient analysis by using modern libraries for the general solution of Partial Differential Equations (PDEs). Typical examples are the MOOSE project and the various activities dedicated to the use of OpenFOAM or COMSOL as multiphysics libraries for reactor applications. The use of general PDE libraries allows for some additional flexibility compared to legacy codes. In particular, one can make use of multiple unstructured meshes for modelling different phenomena (e.g., neutronics and thermal-mechanics), and rely on available mesh-to-mesh projection algorithms for consistently transferring information from one mesh to the other. In terms of neutronic feedbacks from thermal deformations, this paves the way towards the use of a detailed thermal-mechanical modeling. The resulting local displacement field can then be projected to a mesh used for neutronics, interpolated to its points, and used to deform such mesh for a direct prediction of feedbacks based on first principles. This approach is expected to improve predictions in terms of variation of core boundaries and relative position of core and control rods. Some additional improvement may come from a local (instead of core- or zone-averaged) parametrization of cross sections.

A preliminary work from the authors [2] used this approach to predict the thermal deformation feedbacks in the European Sodium Fast Reactor (ESFR). Expansions were based on a simple homogenized thermal-mechanical model of the core, without details about the structure of diagrid and assemblies. The displacement field was then projected on a second mesh used for solving neutron diffusion equations, and used to deform such mesh. Cross-sections were parametrized based on local axial and radial expansions. The results showed that a detailed modelling of the axial fuel expansion could result in 50% lower feedbacks for fuel expansion if compared to approaches based on uniform expansions calculated from average temperatures.

A similar development effort has been carried out at the Argonne National Laboratory based on the SHARP toolkit, which allows to couple advanced codes for thermal-hydraulics, neutronics and thermal-mechanics. It was used in [3] to achieve a detailed simulation of expansion reactivity effects in a SFR mini-core.

Following the promising results of the mentioned studies, this paper discusses an extension of the OpenFOAM-based methodology developed in [2] to include a detailed thermal-mechanic model, as well as more accurate neutronics models. In particular, a thermal-mechanic model is developed where each single assembly is modelled separately and the contact between assemblies is modelled using a dedicated contact boundary condition[1]. As far as the neutronics is concerned, three main possibilities are investigated, namely: 1) standard diffusion/SP3 solvers; 2) a newly implemented discrete ordinate solver; and 3) the use of the Serpent Monte Carlo code via application of its multiphysics interface. The recent ESFR-SMART design [4] is used as a test case.

The paper is organized as follows. Section 2 describes the developed methodologies. Section 3 discusses the obtained results. Finally, the conclusions of the work are reported in Section 4.

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[1] *It is worth recalling that assemblies in SFRs are encased in hexagonal metallic wrappers, which motivates (and justify) an individual assembly representation.*

2. MODELLING DETAILS

The OpenFOAM-based GeN-Foam software [5] was employed as basis for the developed model. The following paragraphs describe the details and the new developments necessary for the thermal-mechanic and neutronic modeling of the ESFR, as well as a discussion about the coupling between them. With regards to the thermal-hydraulic modelling, the standard porous-medium[2] thermal-hydraulic sub-solver of GeN-Foam has been used. The details of the thermal-hydraulic modelling are not described here for brevity, but additional information can be found in [5,6,7].

2.1. Thermal-mechanic modelling

The thermal-mechanic model is based on linear-elasticity theory, including thermal stresses and deformations. As mentioned, the assemblies are modelled in this work as separate bodies. The contact between them is simulated by employing a special boundary condition that was recently developed for the OpenFOAM-based OFFBEAT fuel-behavior solver [8,9]. This boundary condition employs the OpenFOAM Arbitrary Mesh Interface in order to locally evaluate the distance between 2 neighboring and coupled boundary patches. It then applies a contact pressure in case of compenetration of the two bodies. In the current work, this contact boundary condition has been slightly modified to include the possibility of an offset in the contact. If an offset is used, the contact will happen when the distance between two opposing faces becomes lower than the offset. This allows simulating spacers without having to explicitly represent them in the mesh. A fixed displacement or a fixed traction boundary conditions can be applied on the external boundaries of the core in order to take into account the effect of a core restraint system.

The geometry and mesh employed for the thermal-mechanic modelling of the ESFR-SMART core are reported in Figure 1. For this preliminary work, a coarse triangular extruded mesh is employed and the assemblies are modelled as full bodies.

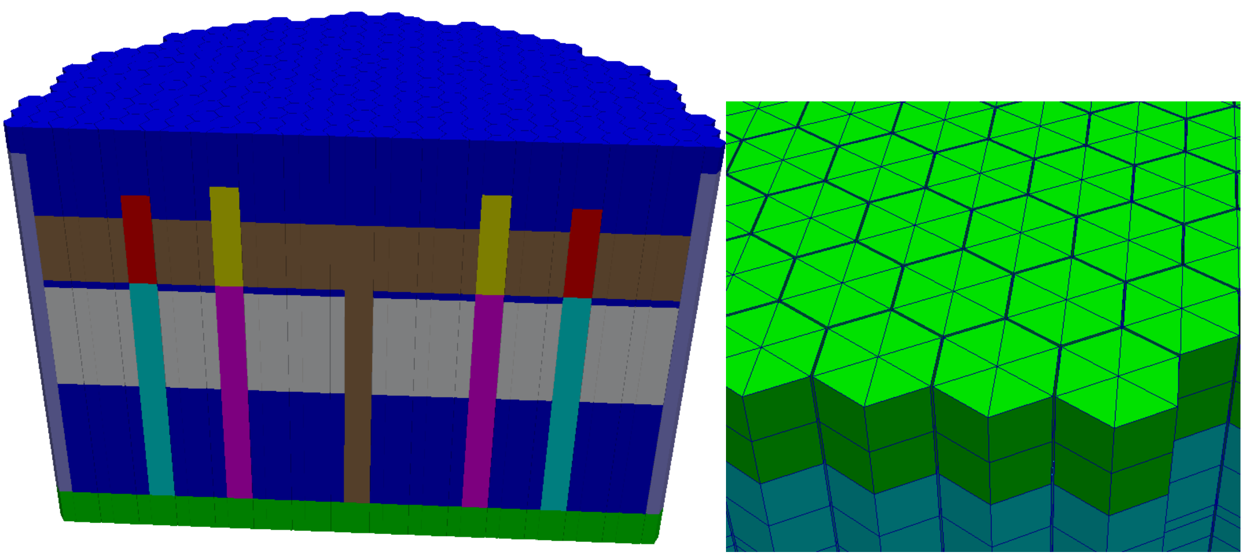

Figure 1. Geometry and mesh employed for an assembly wise thermal-mechanic simulation of the ESFR-SMART core

[2] *Porous-medium modelling is a coarse-grained version of a standard RANS modelling, and it can be considered as a generalization of a sub-channel approach. The geometric details of the fuel pins are not modelled explicitly, but accounted for via dedicated source terms in the energy and momentum equations.*

2.2. Neutronic modelling

The neutronic modelling of a deformed core with explicit representation of the assemblies is non-trivial. First of all, one needs a numerical scheme that allows modelling deformed geometries. In addition, a method is necessary that can capture the effect of neutronically thin regions like the inter-assembly gaps, and of changing volume fractions in the assemblies. One can follow two main routes, Monte Carlo or deterministic.

2.2.1 Monte Carlo modelling

Using a Monte Carlo approach for neutronics is consistent with the high-fidelity modelling employed for the thermal-mechanics, though not allowing (not routinely at least) for transient analyses. In terms of Monte Carlo simulations, there are two main possibilities, namely: with homogeneous or heterogeneous representation of the assemblies.

A simulation based on a homogeneous representation of the assemblies can be performed using the multiphysics interface of the Serpent Monte Carlo code. In this work, a neutronic mesh has been used that is identical to the mesh for the thermal-mechanics. This mesh has been deformed using the displacement field calculated by the thermal-mechanic modelling and fed to the multiphysics interface of Serpent. This interface is capable of identifying the cell where each neutron interaction happens, and to associate it to a homogenized material. Unfortunately, the current structure of the Serpent multi-physics interface does not allow for a selective variation of the fraction of the materials filling the assemblies. As a consequence, one can only simulate a deformation that does not alter the volume of the assemblies (such as a radial expansion driven by the diagrid). Any thermal deformation that leads to a change of the cross-sectional area of the assemblies would require the possibility to modify the sodium fraction accordingly.

For what concerns the possibility of a heterogeneous representation of the assemblies, there are two possible approaches. One would be to employ the Serpent multi-physics interface, but applied to a detailed pin-by-pin OpenFOAM geometry. The other would be to use the standard combinatorial geometry of Serpent, but modifying the tracking routine of the Monte Carlo code so that a deformation field is added to the interaction points. Both approaches are currently under investigation, but their implementation if beyond the scope of the present work.

2.2.2 Deterministic modelling

A deterministic modelling approach is typically used for transient analyses in the fast reactor community. Most of the time, this is achieved via a diffusion analysis, with cross-sections parametrized with respect to, among others, radial and axial fuel expansion. However, the choice of separately modelling assemblies and gaps entails some difficulties using these models. The first problem is that the gaps are neutronically thin, and thus not suited for a diffusion approximation. The second problem is that a radial expansion driven by the diagrid would result in a change in the gap size. In a traditional homogeneous representation of assemblies and gaps, this is taken into account via a parametrization of the cross-sections. With an explicit representation of the gaps, the neutronic solver itself has to be able to model the spatial and spectral effect of gaps with variable dimensions. For this reason, a discrete ordinate solver has been included in GeN-Foam. The solver is an adaptation and extension of the work from Refs. [10,11]. Beside adaptation and coupling, the only significant extension compared to the work in Refs. [10,11] is the added possibility to assign different cross-sections to different regions of the mesh.

In the current paper, three energy structures have been employed: a standard 24 group structure [2]; a refined 46-group structure obtained by adding a group boundary at the mid-points of the 24-group structure; and a simple 1-group structure. For the discrete ordinate simulations, a 128-direction quadrature set has been employed [12]. A finer mesh compared to the thermal-mechanic one has been used for the analyses (Figure 2).

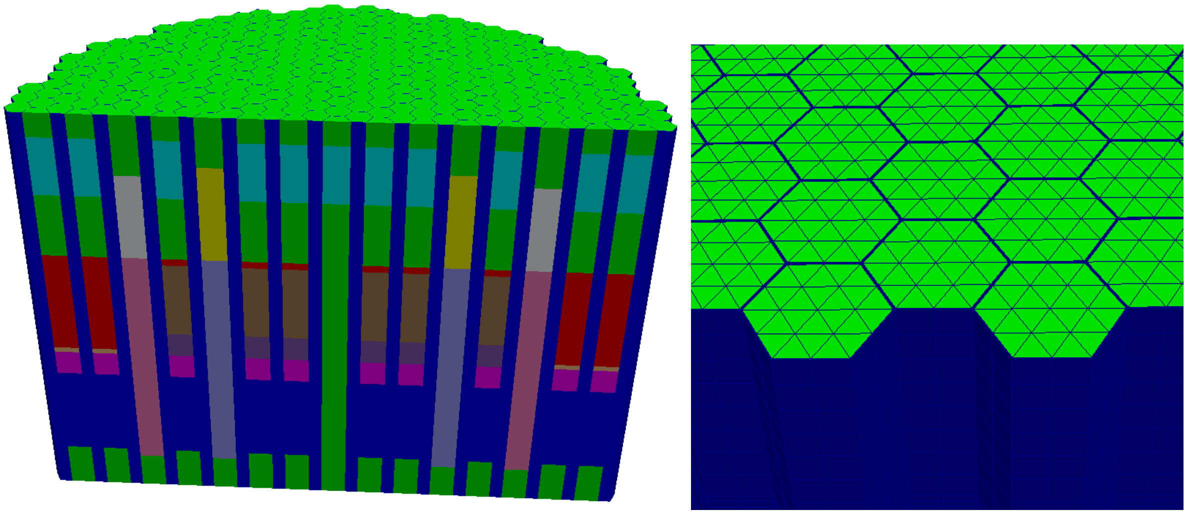

Figure 2. Geometry and mesh employed for an assembly-wise neutronic simulation of the ESFR-SMART core

2.3. Coupling neutronics and thermal-mechanics

Neutronics and thermal-mechanics are implicitly coupled via fixed-point iterations and quantities are exchanged between the two meshes using existing OpenFOAM projection algorithms. In GeN-Foam, a conservative volume-weighted projection algorithm is normally employed. However, the displacement field is not defined in the inter-assembly gaps of the thermal-mechanic mesh, while these gaps are instead explicitly modelled in the neutronic mesh. This implies that the displacement field projected to the neutronic mesh would be undefined in the inter-assembly gap. An undefined value normally results in a default value of zero in projection algorithms. This has two negative consequences. The first one is that a large radial displacement (e.g., towards the radial periphery of the core) may cause the points belonging to the assembly to overlap with points belonging to the gap. The second is related to the fact that the mesh displacement is achieved by interpolating cell center values to the mesh points. In the points that separate assembly and gap, this would imply a displacement that is averaged between the correct one in the assembly and the zero value in the gap. The result is a significant core distortion (see Figure 3, left).

To obviate this problem, in this work a projection algorithm is employed based on a nearest-point strategy. In this way, each cell of the mesh for neutronics would take a displacement value from the spatially closest cell in the mesh for thermal-mechanics. A cell belonging to the inter-assembly gap in the neutronic mesh would then take its value of displacement from the neighboring assemblies, thus allowing for the points in the gap to naturally "follow" the displacement of the assemblies. Figure 3 (right) shows the results of a displacement field based on a nearest-point algorithm.

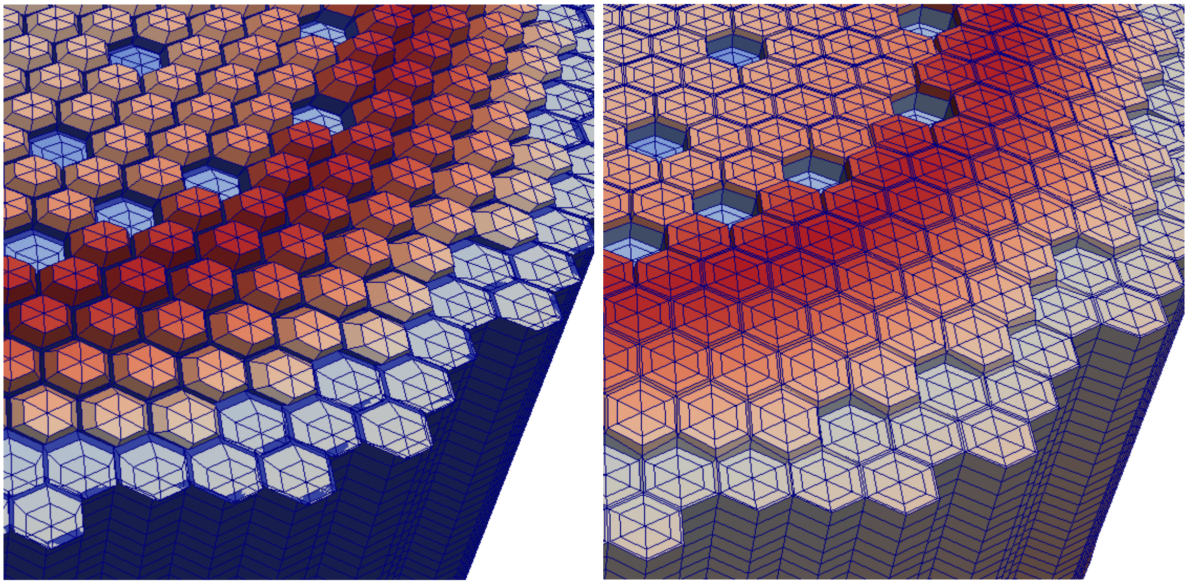

Figure 3. Deformation of the neutronic mesh based on volume-averaged (left) and on a nearest-point (right) projection algorithms. A web-shaped mesh has been used to facilitate the visualization

3. RESULTS

The model described in the previous section allows for a detailed and accurate analysis of the core thermal expansion and related reactivity feedbacks in fast reactors. The model has been tested in the present work using the ESFR-SMART design as case study.

At first, an investigation has been carried out to evaluate the capability of different neutron transport models to reproduce the reactivity effect of an expanded core. To this purpose, a simple case is adopted where the assembly pitch is increased by 5%, while leaving unaltered the assemblies themselves. This allows to employ the coupled Serpent-OpenFOAM model to generate a reference solution.

Table I shows the reactivity decrements that have been calculated using selected combinations of different neutronic models and energy-group structures. The reported values have to be compared with a Serpent-OpenFOAM reference solution of 2608 pcm.

Table I. Reactivity decrements (pcm) due to a 5% increase of the assembly pitch in the ESFR-SMART design

	Diffusion	**SP3**	**SN-P0**	**SN-P1**
1	2904			
24	3293	3154	2219	2510
46	3366			
Reference solution	2608			

Table I shows significantly inaccurate predictions of the diffusion model. As mentioned, this is an expected outcome that results from the unsuitability of diffusion theory to model neutronically thin regions like the inter-assembly gaps. The SP3 model provides slightly improved results, even though its accuracy is still low. A significant improvement is instead obtained using the discrete ordinate (SN) solver, notably when employing an anisotropic (P1) modelling of scattering. The accuracy obtained by using the discrete ordinate model can be considered as satisfactory within the scope of the work. It could be further improved in the future by increasing the number of energy groups, the number of discrete directions, the order of the Legendre expansion in the scattering approximation, and by further refining the mesh[3]. From Table I, one can also observe the importance of a multi-group approach, which allows to capture the spectrum shift caused by a larger gap and a higher sodium-to-fuel ratio in the core. In this sense, 24 groups appear to be enough for modelling the essential features of such an effect, though some changes in the results are still observed when using 46 groups.

Based on the results reported in Table I, the SN-P1 model has been employed to evaluate the effect of core flowering, and of a hypothetical core compaction, in the ESFR-SMART design. To this purpose, two coupled (thermal-hydraulics, thermal-mechanics, neutronics) steady-state simulations have been performed at the nominal ESFR-SMART power. The flowering has been obtained in the standard case where the assemblies are free to move radially at the top of the core. The core compaction is obtained by constraining the axial top of the assemblies (like in the EBR-I reactor). The results of the simulations are reported in Figure 4.

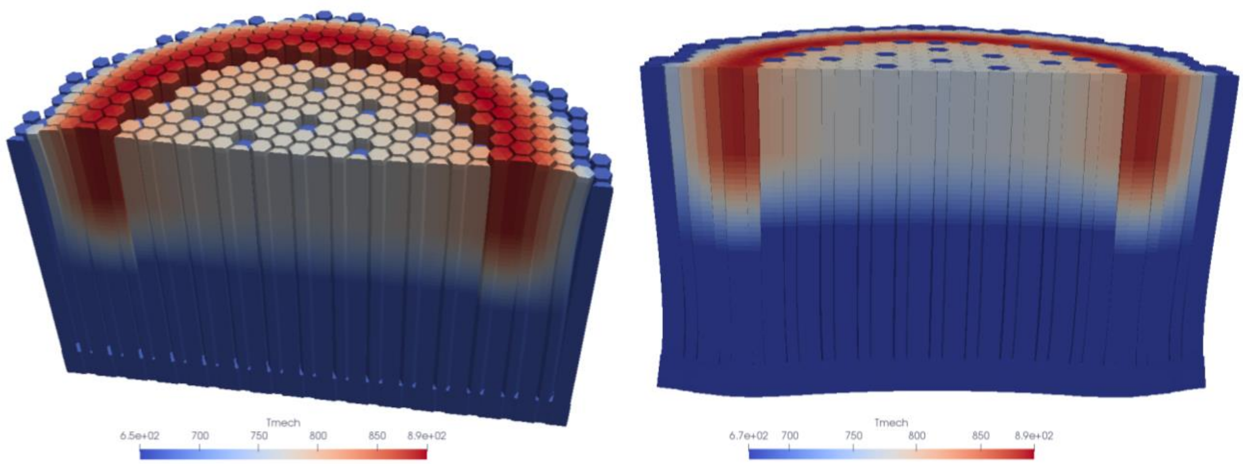

Figure 4. Core flowering (left) and core compaction (right) for the ESFR-SMART design (magnified)

In these simulations, the assemblies can freely expand and change their cross-sectional area. This determines a variation in the sodium quantity, which has been accounted for by parametrizing the cross-sections with respect to the local intra-assembly expansion. In order to isolate the effect of radial expansion, no other parameter has been used for cross-section parametrization.

The reactivity effect of core flowering and compaction has been calculated by comparing the obtained multiplication factors with that of the hot zero power reactor state (isothermal at inlet sodium temperature). In addition, only the calculated radial displacement is employed to deform the neutronic mesh, while the axial component is set to zero. As expected, the core compaction leads to a positive reactivity insertion. The calculated value is 69.7 pcm. More surprisingly, the core flowering also determines a small positive reactivity insertion of 31.3 pcm. An explanation for this is the bending of the assemblies away from the control rods, notably at the boundary between inner and outer core. This bending is due to the strong power

[3] *A full mesh convergence has been observed for the diffusion and SP3 models using the mesh shown in Figure 2. Slightly improved results are instead expected by further refining the mesh when using the discrete ordinate model. However, a further refined mesh would require computational resources that are currently unavailable to the authors. Further studies in this sense are planned.*

(and temperature) gradient in this interface region and can be better observed in Fig. 5, where the behavior of the single assemblies has been highlighted by representing one assembly every two, and by strongly magnifying the distortion. This explanation has been preliminary confirmed by performing the simulations without followers and control rods. In particular, followers have been substituted with fuel and control rods with the upper plenum. With this core configuration, the core flowering has resulted in a slightly negative reactivity reduction of few pcm. The small value of the reactivity change in case of flowering is consistent with a calculated overall flowering of ~4 mm at the top of the core. This corresponds to less than 2mm on average, which translate into a reactivity decrement of 3.5 pcm (estimated using the ~2600 pcm for a 5% expansion calculated above).

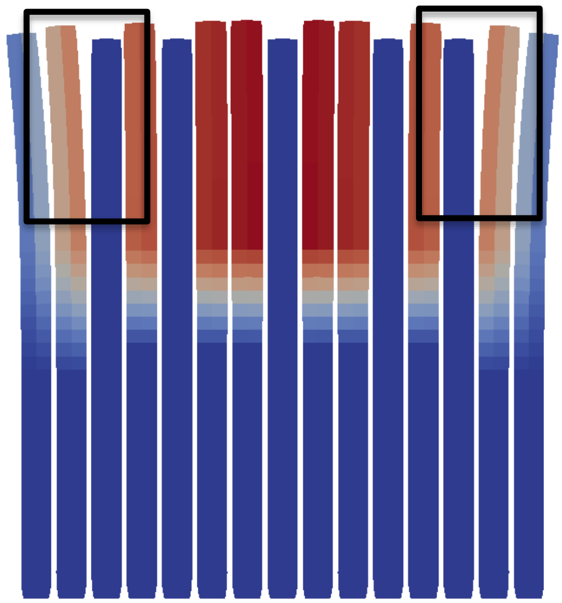

Figure 5. Core flowering (magnified) at the core symmetry plane. Only one assembly every two is represented

4. DISCUSSION AND CONCLUSIONS

The paper has presented a GeN-Foam based model for the detailed analysis of expansion-related reactivity feedbacks in fast reactors. The model implicitly couples three sub-solvers for coarse-mesh thermal-hydraulics, thermal-mechanics and neutronics. The thermal-mechanic sub-solver allows for an explicit treatment of separate assemblies and the contact between them. The neutronic sub-solver allows for mesh deformations and can rely on three different neutronic models, namely: diffusion, SP3, and discrete ordinates. In addition, a coupling with Serpent is possible by using the standard Serpent multiphysics interface.

The results obtained in the paper demonstrate the capability of the model to predict complex thermal-mechanic phenomena like the core flowering and the core compaction. However, an accurate prediction of the related reactivity feedbacks requires computationally expensive models like Monte Carlo or discrete ordinate models. As a matter of fact, a Monte Carlo simulations can currently be performed only for cases where the assemblies themselves are not distorted. When simulating core flowering and compaction, this limits the choice of the neutronic model to the discrete ordinate one. Two coupled thermal-hydraulic/thermal-mechanic/discrete-ordinates solutions have allowed to preliminary quantify the reactivity impact of these phenomena in the ESFR-SMART design. As a result, core compaction results in a significant reactivity increment. Core flowering itself has instead a negligible impact on reactivity, but the bending of the assemblies appears to have a significant and positive effect.

Future work on the subject will focus first and foremost on verification and validation. Preliminary calculations for the Godiva experiment have confirmed the capability of the SN model to reproduce experimental results (~100 pcm discrepancy using 128 directions, 23 energy groups, and a p3 approximations for scattering). In addition, the results reported in the paper shows a very good comparison with the results obtained using a reference Serpent-OpenFOAM model. Nonetheless, further efforts are required to verify the capability of the model to deal with complex simulations where the reactivity feedback come both from a changing inter-assembly gap, and from a distortion of the assembly. The Phenix end-of-life tests are being considered for this purpose.

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